

DILEEPA D

Aspiring Python / Backend Developer | Data Science | AI/ML | Generative AI

Bengaluru, India | 9108159450 | dileepad2005@gmail.com
LinkedIn | GitHub

PROFESSIONAL SUMMARY

Final-year Computer Engineering student pursuing a B.E. with a CGPA of 8.2/10, graduating in June 2027. Hands-on experience building Python-based applications, deep learning systems, data-analysis solutions, and Generative AI workflows. Comfortable with backend logic, relational databases, data pipelines, machine learning, NLP, RAG, and LangChain. Strong foundation in Data Structures & Algorithms, Git/GitHub, and end-to-end project delivery from data processing to model integration. Seeking opportunities in Python, Backend, Data Science, AI/ML, or Generative AI roles.

TECHNICAL SKILLS

Programming: Python, SQL, HTML, CSS, JavaScript, Data Structures & Algorithms (DSA)

Databases: MySQL, PostgreSQL, MongoDB/BSON fundamentals

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization

Machine Learning: Scikit-learn, Linear Regression, Logistic Regression, Polynomial Regression, KNN, Decision Trees, Naive Bayes, K-Means, PCA, SVD, Model Evaluation

Deep Learning & Computer Vision: TensorFlow, Keras, CNN, OpenCV, DeepFace, Image Classification, Facial Emotion Recognition

NLP: TF-IDF, Word2Vec, Gensim, Sentiment Analysis, Text Classification

Generative AI: RAG (Retrieval-Augmented Generation), LangChain, LLMs, Embeddings, Semantic Search, Prompt Engineering, Vector Search, Sentence Transformers

Big Data: Apache Hadoop, Apache Spark, Distributed Data Processing

Backend & Deployment: Flask, Gradio

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

Real-Time Face Emotion Recognition System

Oct 2024 – Nov 2024

- Developed a real-time facial emotion recognition system in Python using Convolutional Neural Networks (CNN) and OpenCV to classify human emotions from live facial expressions.
- Trained and evaluated a deep learning image-classification model and integrated it into a functioning end-to-end application.
- Technologies:** Python, TensorFlow, OpenCV, CNN, Deep Learning

Netflix Movies & TV Shows Data Analysis

- Analyzed a dataset of 8,000+ Netflix titles using Python, Pandas, and NumPy to identify content trends and release patterns.
- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) to prepare raw data for downstream modeling.
- Built a content-based recommendation system and communicated findings through visualizations created with Matplotlib and Seaborn.
- Technologies:** Python, Pandas, NumPy, Matplotlib, Seaborn

Big Data Social Media Sentiment Analysis & User Behavior Mining

- Analyzed large-scale social media data to identify sentiment patterns and user behavior using NLP and Big Data techniques.
- Applied TF-IDF and Naive Bayes-based classification for sentiment analysis and explored distributed processing with Hadoop and Apache Spark.
- Technologies:** Python, Hadoop, Spark, NLP, TF-IDF, Naive Bayes, Pandas

RAG / LangChain AI Pipeline

- Studied and implemented Retrieval-Augmented Generation (RAG) concepts for retrieving relevant information from documents and providing context-aware LLM responses.
- Worked with LangChain, embeddings, semantic search, and vector-based retrieval as components of an AI/RAG

pipeline.

- **Technologies:** Python, RAG, LangChain, LLMs, Embeddings, Sentence Transformers

EDUCATION

Don Bosco Institute of Technology — Bachelor of Engineering (B.E.), Computer Science and Engineering Aug 2023 – Jun 2027 (Expected)

CGPA: **8.2 / 10**

CERTIFICATIONS

- Introduction to Generative AI — SkillUp
- Software Engineering Job Simulation — JPMorgan Chase, via Forage
- BASH Training and RDBMS PostgreSQL Training — Edupyramids

ADDITIONAL COURSEWORK

- Software Project Management — NPTEL, IIT Kharagpur (Ministry of Education, Government of India)
- Operating System Fundamentals — NPTEL, IIT Kharagpur (Ministry of Education, Government of India)
- Data Structures & Algorithms, Database Management Systems, Operating Systems, Software Engineering, Big Data, Machine Learning, NLP